Mohammad Sefid — Curriculum Vitae

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mohammad-sefid

GitHub QR code.png

Research Summary

- Master's level research in RIS-aided Integrated Sensing and Communication (ISAC) systems with Cooperative NOMA via Energy Harvesting from SWIPT operation and optimization with Deep Reinforcement Learning (DRL), Alternating Optimization (AO), and Metaheuristic techniques

Research Experience

- **K. N. Toosi University of Technology** — **Tehran, Iran**
  Graduate Researcher — RIS-Assisted ISAC Systems — *2023–2026*

  Research projects:
  - **Performance improvement of RIS-assisted ISAC systems**
  - **Machine Learning** (Code)
  - **Numerical Optimization** (Code)
  - **Digital Signal Processing** (Code)
  - **Adaptive Filters** (Code)

Publications

- IRS-Assisted Cooperative NOMA with SWIPT in ISAC Systems: A Comparison of Half-Duplex and Full-Duplex Modes [In progress]

Teaching Experience

- **K. N. Toosi University of Technology** — **Tehran, Iran**
  Teaching Assistant — *Feb 2025- Jan 2026*

  - **Numerical Optimization**
  - **Stochastic Processes**
  - **Estimation Theory**
  - **Information Theory**

  Reviewed and graded the computer-based and theoretical assignments of graduate students. In addition, prepared and provided the correct code and the solution key for the questions.

Education

- **K. N. Toosi University of Technology** — **Tehran, Iran**
  MSc Electrical, Electronics and Communications Engineering — *2023–2026*

  Thesis research: RIS-aided ISAC with cooperative NOMA and SWIPT
  Relevant coursework: Machine Learning, Numerical Optimization

- **University of Applied Science and Technology** — **Iran**
  Bachelor's Degree, Electrical and Electronics Engineering

  Notable Projects:
  - **A Review of Photovoltaic (PV) Technologies: System Components, Architectures, and Future Research**

Directions

- **Arak University of Technology** **Iran**
  Associate Degree, Electrical and Electronics Engineering

Technical Skills

- **Programming:** Python | MATLAB | GitHub
- **Data Analysis:** Optimization | Machine Learning
- **Writing Tools:** LaTeX| Microsoft Office

Conference Presentation

- Conference on Electrical and Computer Innovations, K. N. Toosi University of Technology, Feb 2025
  Presented my research on "Applications of STAR RIS in Communication Systems"

Research Interests

- Joint Optimization of Integrated Communication and Sensing (ISAC) Performance in Next-Generation Wireless Networks
- Simultaneous Wireless Information and Power Transfer (SWIPT) for Cooperative Non-Orthogonal Multiple Access (C-NOMA) Systems
- Optimization of Reconfigurable Intelligent Surfaces (RIS) to Enhance Beamforming of Wireless Signals
- Deep Reinforcement Learning (DRL) optimization for adaptive wireless Systems

Industry Experience

- **Manasazan Sadid** **Tehran, Iran**
  Intern *Apr-June 2021*

  - Gained practical experience in the planning and design of renewable energy power plants
  - Supported the installation of energy systems, including off-grid solar power plants
  - Developed understanding of installation project management and logistics coordination

- **Zamyad Co.** **Tehran, Iran**
  Intern *Jul-Sep 2018*

  - Developed knowledge of automotive electrical and electronic systems
  - Provided electrical and electronic support for quality control laboratories and manufacturing lines
  - Evaluated safety measures and contributed to the installation of grounding (earthing) systems for technical lines

- **Ariosunshade** **Tehran, Iran**
  Technical content creator *Sep 2017-Mar 2018*

  - Translated technical energy-related articles from English into Persian
  - Conducted literature review to identify state-of-the-art advancements in the energy sector

Additional

- Half Marathon Runner
- Recreational Cyclist
- Regular Workout